Teaching Note: Explainable Artificial Intelligence

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Date: April 2026

1.1 Introduction

Artificial intelligence (AI) models are increasingly being integrated by business experts into operational decision-making, given their strength in prediction and optimization over traditional techniques. Today, AI models are used across many firms and industries for pricing (Caro and Sáez de Tejada Cuenca 2023, Kawaguchi 2021), customer marketing (Lambrecht and Tucker 2019), risk scoring (Kelley et al. 2022), forecasting (Baardman et al. 2018), supply chain management (Mišić and Perakis 2020), etc. The higher accuracy of AI models does not always lead to outright adoption of AI recommendations though, business experts are often reluctant to adhere, despite the AI providing superior (although imperfect) recommendations, a phenomenon referred to as algorithm aversion (Dietvorst et al. 2018). While more accurate, AI models are often less explainable than traditional statistical techniques, resulting in many AI models being referred to as "black-box." To address the reduced explainability of AI models and potential algorithm aversion, computer scientists have developed several explainable AI (XAI) approaches (Langer et al. 2021). XAI approaches produce details or rationale to make an AI model's functioning easier to understand. XAI approaches include interpretable models (also referred to as "glass-box") such as logistic regression or explainable boosting machine, which by design offer explanations (often parameters) with predictions; and model agnostic post-hoc techniques, such as decision trees, and SHAP values, which are applied after the AI model has been developed to generate an explanation (Arrieta et al. 2020).

Beyond the use of XAI visualizations to improve AI recommendation adherence in operational decisions, XAI visualizations can help uncover possible biases within a model. Current and forthcoming AI regulations including the European Union (EU) AI Act (Artificial Intelligence Act, 2021), Singapore's Personal Data Protection Act (Personal Data Protection Act 2012, 2020), and the United States (US) Blueprint for an AI Bill of Rights (The Office of Science and Technology Policy, 2022), all highlight the need for explainable algorithms to reduce ethical issues, like bias. In this light, the US Blueprint for an AI Bill of Rights states that "…automated systems should provide explanations," and that explanations should be "tailored to the target." It suggests that "tailoring should be assessed (e.g., via user experience research)."

1.2 XAI Approach

According to Explainable AI (XAI) methods, machine learning models can be categorized into three broad groups in terms of inherent explainability or interpretability (Arrieta et al. 2020). 1)

Interpretable models utilize the natural explainability of simpler models, including “glass-box” models such as logistic and linear regression, General Additive Models, and Naïve Bayes Classifier; and new interpretable ML models, like Explainable Boosting Machine (EBM). 2) Model-agnostic methods which are compatible with several AI models (including black box models) and usually applied post-hoc, such as decision trees, feature importance, and SHAP values. 3) Lastly, example-based explanations which take samples from the dataset to visualize the system’s behaviour under what-if scenarios, such as counterfactuals.

The XAI approach is but one of several attributes of XAI visualizations. Three additional key attributes are the number of features in the visualization, the applicant aggregation level, and the model used to generate the prediction and resulting XAI visualization.

Feature importance (FI) is the practice of assigning a normalized score, often between 0 to 1, to each feature based on how powerful it is in predicting the target variable. It is computed by comparing the difference in model performance metrics using the original values of the features with a shuffled value of the same feature. In essence, a feature is considered more important if the difference in error is larger than those of other features (Saarela and Jauhiainen 2021). A 1-sided FI shows features that either all have a positive impact on the prediction or all have a negative impact on the prediction. See Figure 1, Panel A for an example. A 2-sided FI method operates in a similar vein to its 1-sided counterpart, while showing both positive and negative values. See Figure 1, Panel B for an example. In a binary classification model, a positive value implies that the feature influences the active outcome (i.e., in our setting, default) while a negative value implies the feature influences the passive outcome (i.e., no default).

Shapley values, also known as SHAP values, are another model-agnostic explainable technique often used to interpret results from black-box models like XGBoost or neural networks, based in game theory (Lundberg and Lee 2017). The SHAP value of a feature is the average impact it has on the model's prediction, considering all possible ways it can interact with other features. SHAP visualizations differ depending on whether the explanation is displayed at the individual applicant or global level (referred to as applicant aggregation level, see §3.3); we test both visualization types. See Figure 1, Panel C, for the group level example.

Decision Trees (DTs) encompass a host of algorithms on their own but can also be used to visualize interactions between features, especially when the relationship between model outcomes and features is non-linear (Sagi and Rokach 2020). Paths from a tree root node branch out into multiple nodes of many layers using the Gini index or information gain and thus can also be used to represent feature importance. See Figure 1, Panel D.

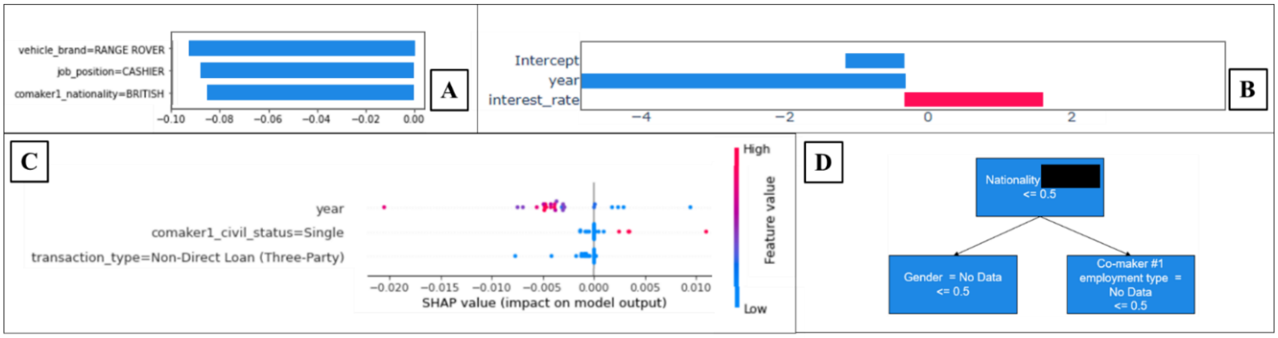

Figure 1. XAI Approaches (Panel A: 1-sided FI, B: 2-sided FI, C: SHAP Values, D: DT)

1.3 Number of Features

Another factor for consideration in XAI visualization is the amount of information provided, or the number of features in the visualization. An average person's processing capability may be unable to absorb and manage high levels of information and could result in information overload or poor information retrieval (Lopatovska and Arapakis 2011). While graphs and visualizations may be used to mitigate these adverse effects, information overload is known to negatively affect managerial decision quality and decision-making outcomes in high-complexity settings (Chan 2001, Falschunger et al. 2016); therefore, finding the preferred amount of information could help drive greater use of the XAI visualizations. Often, XAI visualizations with the top three, six, and nine features according to model results were generated. While showing less information may be less overwhelming for some users, not enough information about the model could work against its preference; for example Foerster et al. (2020) found that longer explanations are frequently described as concrete (a positive attribute) versus shorter explanations. As an example, the XAI visualizations in Figure 1 each have three features; for the DT (Panel D) we count the number of leaves (boxes) as features.

1.4 Applicant aggregation level

XAI approaches can be applied to explain model results at either an individual or group level (Schrouff et al 2021). At the global aggregation level, features represent importance for the entire model, whereas at the individual (or local) level, features show the contribution for a particular observation (in our setting, a single lending applicant). Initial experimental work set in the online dating industry found that global and local XAI did not differ in how they impacted a user's trust in an AI system (Ahn et al. 2024). See Figure 2, Panel A for an example of a 1-sided FI visualization at the individual level, and Panel B for the global level.

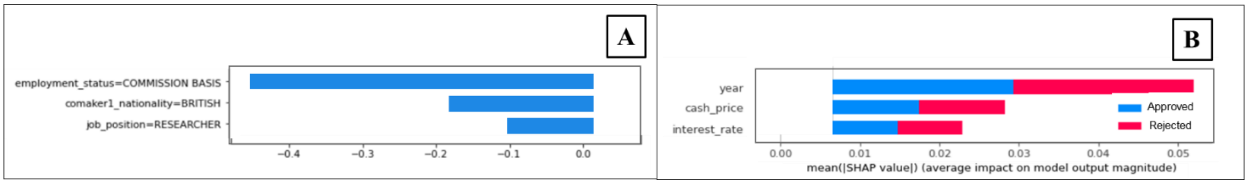

Figure 2. Examples of Applicant Aggregation Level (Panel A: Individual, B: Global)

1.5 Model type

The interpretability spectrum ranges between black-box and glass-box models. Black-box, usually machine learning models, are characterized by higher performance measures such as accuracy, precision, and recall, but do so by forfeiting ease of interpretability; providing predictions but little to no model parameters in the output (Rai 2020). Glass-box models, by design, supply model components that are easily interpretable, and can range from traditional statistical techniques to more recent interpretable machine learning.

The AI outcome (positive or negative) can also impact an AI user's preferences, given that affected parties have been found to have different visualization preferences in positive and negative outcome situations (Lu et al. 2020).

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